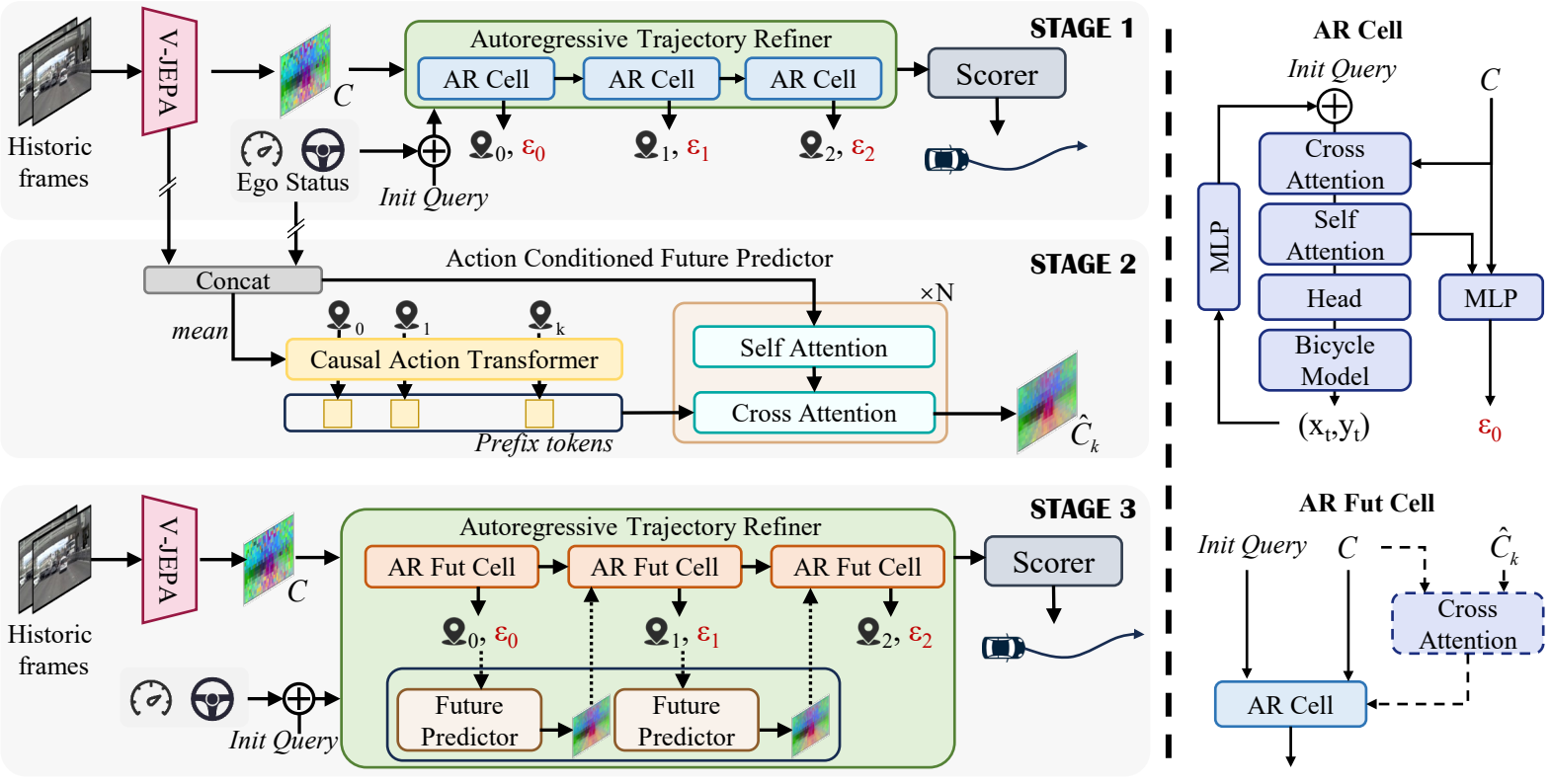

STAGE 1
Historic frames
V-JEPA
C
Autoregressive Trajectory Refiner
AR Cell
AR Cell
AR Cell
Scorer
Ego Status
Init Query
0, ε0
1, ε1
2, ε2
STAGE 2
Action Conditioned Future Predictor
Concat
mean
0
1
k
Causal Action Transformer
Prefix tokens
×N
Self Attention
Cross Attention
Ĉk
STAGE 3
Historic frames
V-JEPA
C
Autoregressive Trajectory Refiner
AR Fut Cell
AR Fut Cell
AR Fut Cell
Scorer
0, ε0
1, ε1
2, ε2
Future Predictor
Future Predictor
Init Query
AR Cell
Init Query
C
Cross Attention
Self Attention
Head
Bicycle Model
MLP
MLP
(xt,yt)
ε0
AR Fut Cell
Init Query
C
Ĉk
Cross Attention
AR Cell